WeCare

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1 Introduction

Delayed detection, alert fatigue, and context loss are the three core problems WeCare is built to solve. WeCare is a real-time patient monitoring system designed for nurses and care coordinators in elderly care facilities, providing immediate situational awareness without overwhelming staff with noise.

These problems aren't hypothetical. The UK care-home sector serves over 400000 residents [1], many living with dementia and multiple chronic conditions [2], yet most facilities still rely on periodic manual checks. A fall or a drop in oxygen levels can go undetected for critical minutes between rounds, minutes that can prove fatal in elderly patients [3]. Meanwhile, systems that surface every minor fluctuation train staff to ignore them entirely [4]. And a raw number like a heart rate of 48 is clinically meaningless without a baseline and trend to compare it against.

WeCare addresses all three simultaneously. Streaming rules fire only on clinically significant events, severity routing separates urgent from advisory signals, and patient-specific baselines provide immediate context at a glance.

2 Implementation & System Flow

WeCare's architecture was designed with one constraint: every technology choice had to directly serve the three clinical problems outlined above. WeCare processes data through a four-stage pipeline:

Ingestion (Kafka): Continuous telemetry from patient sensors is pushed directly to Apache Kafka. Sending sensor data directly to a backend risks the system becoming unstable when data arrives faster than it can be processed. Kafka sits in between, acting as a buffer that absorbs incoming data and releases it at a manageable pace, keeping the system stable even during busy periods.

Stream Processing (Flink): This is where WeCare's core innovation lives. A simple script could flag a single abnormal reading. However, that is exactly the kind of noise

that causes alert fatigue. Flink instead calculates 10-second rolling averages for heart rate, meaning alerts only fire when a genuine trend is detected rather than a one-off sensor reading. This single decision directly addresses all three motivating problems: real deterioration is caught earlier through continuous analysis, alert fatigue is reduced because only clinically meaningful changes trigger a response, and context loss is solved because nurses get notified from a trend rather than a raw isolated number that could mean nothing on its own.

Alert Routing (RabbitMQ): Once Flink identifies an anomaly, it publishes the alert to RabbitMQ. Kafka is well-suited for handling large volumes of incoming data, but it is not designed to guarantee that each alert is delivered reliably. RabbitMQ handles this by routing each alert to a dedicated queue (**nurse-urgent-alerts** vs `nurse-warnings`), so a critical fall alert is never lost or buried alongside a minor advisory.

Dual-State Storage (Redis & DynamoDB): The system uses a CQRS-style (Command Query Responsibility Segregation) [5] pattern to solve two distinct needs with two distinct tools. Redis provides very fast read performance that the live dashboard depends on, giving nurses an up-to-date view of every patient at a glance. At the same time, every alert is permanently recorded in DynamoDB using logic that prevents duplicate entries, creating a reliable CQC-compliant audit trail [6]. Using a single database for both would mean compromising on either speed or reliability, so two purpose-fit tools are used instead.

A diagram of WeCare's data flow can be observed in Figure 1.

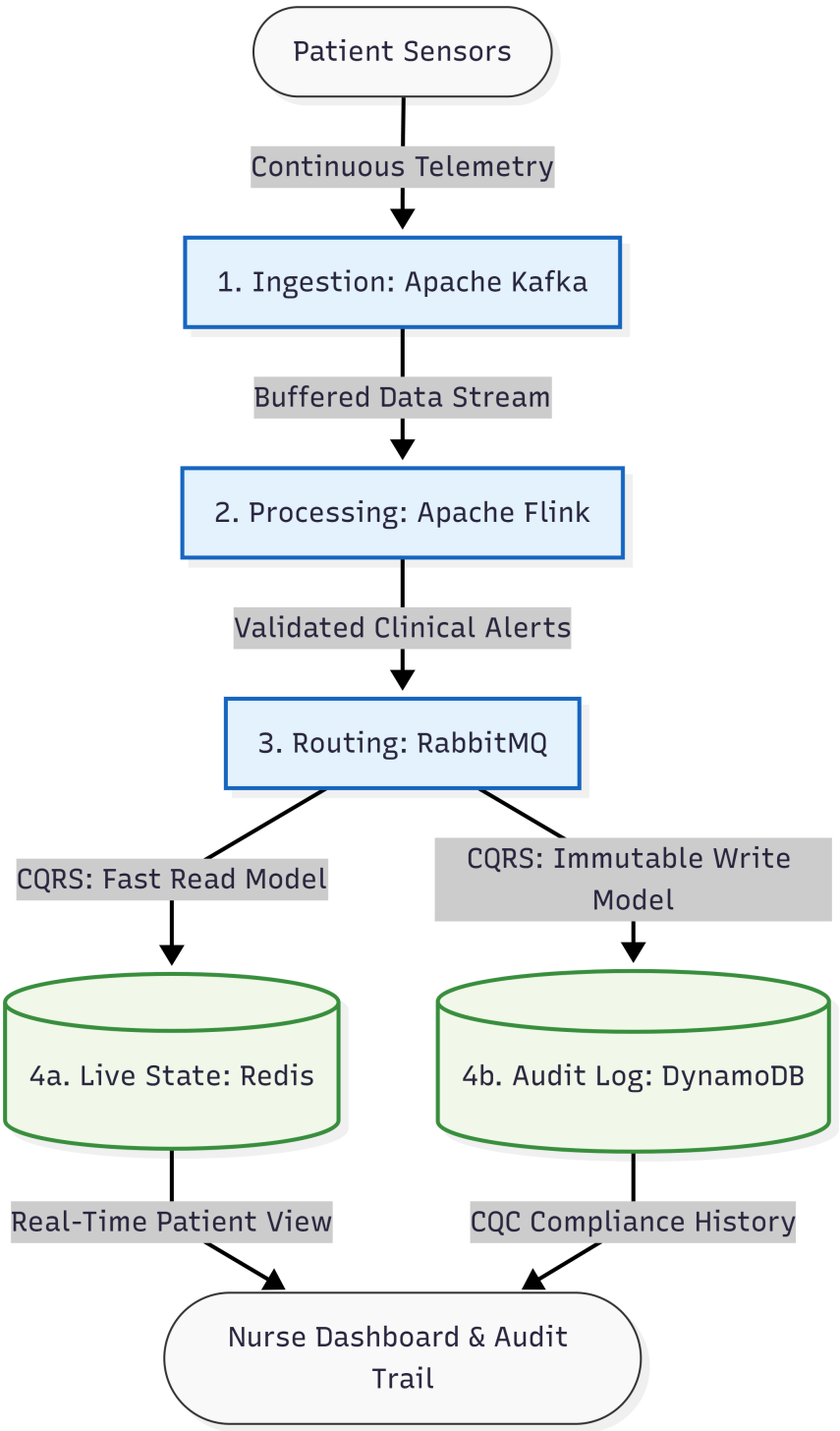

Figure 1: WeCare data flow

3 Engineering Scope & Delivery

WeCare was developed as an end-to-end Proof of Concept (PoC) to demonstrate that a real-time clinical alerting pipeline can work reliably. In healthcare telemetry, the primary engineering risk is the stability of the streaming pipeline. Therefore, the scope of this PoC was limited to integrating message brokers and stream processors, demonstrating that an event-driven architecture can mitigate clinical alert fatigue in real time.

The supported patient warnings are displayed in Table 1. These alerts were chosen as they rank amongst the most common and critical types of incidents within care homes [7] [8].

Rule	**Trigger**	**Severity**
VITALS_OXYGEN_LOW	SpO2 $< 90\%$	URGENT
GEOFENCE_EXIT_GATE	Location = Exit Gate	URGENT
FALL_DETECTED	Movement = Fall Detected	URGENT
PATIENT_RECOVERED	Movement = Recovered	URGENT
HEART_RATE_TREND_10S	10 s avg HR < 50 or > 120 bpm	WARNING/URGENT

Table 1: Flink anomaly detection rules and severity routing

To ensure the pipeline could be properly tested without waiting for statistical coincidence in synthetic data, the telemetry simulator developed to generate data includes an interactive fault injection interface. Developers can manually trigger any critical scenario, for example, **fall 1** or **crash 3**, and immediately watch the alert travel through Kafka, get processed by Flink, routed through RabbitMQ, and reflected on the dashboard in real time. This was a deliberate engineering decision: a monitoring system that cannot be reliably tested cannot be trusted in a clinical setting.

What remains out of scope for this PoC includes patient authentication, role-based access control, and integration with real sensor hardware. These are natural next steps for a production system, but were intentionally set aside to keep the focus on proving the core data pipeline.

4 AI Collaboration

Claude and GitHub Copilot were used throughout the development of WeCare, though in distinct ways. Claude was primarily consulted at the architecture level, helping evaluate technology choices and think through how components like Flink, RabbitMQ and Redis should interact to meet the clinical requirements. Copilot assisted at the code level, accelerating the implementation of boilerplate logic and the dashboard UI.

However, all key decisions remained mine. AI was used as a teacher/instructor and accelerator rather than a replacement for engineering judgment. For example, the decision to use Flink's windowing over a simpler scripting approach came from understanding the clinical problem first, with AI helping validate and implement that direction. This allowed the limited development time to be spent on the parts of the system that required the most thought, rather than on repetitive implementation work.

5 Build Instructions

Instructions to build the project can be found in the repository's README.md.

References

[1] B. Hanratty, D. Stow, D. Collingridge Moore, N. K. Valtorta, and F. Matthews, "Loneliness as a risk factor for care home admission in the english longitudinal study of ageing," *Age and ageing*, vol. 47, no. 6, pp. 896–900, 2018.

[2] J. Bisla, M. Calem, A. Begum, and R. Stewart, *Have we forgotten about dementia in care homes? the importance of maintaining survey research in this sector*, 2011.

[3] C. Downey, S. Chapman, R. Randell, J. Brown, and D. Jayne, "The impact of continuous versus intermittent vital signs monitoring in hospitals: A systematic review and narrative synthesis," *International journal of nursing studies*, vol. 84, pp. 19–27, 2018.

[4] H. R. Anderson et al., "Stats on the desats: Alarm fatigue and the implications for patient safety," *BMJ Open Quality*, vol. 12, no. 3, 2023.

[5] M. Kindson and M. Péter, "A simplified approach to distributed message handling in a cqrs architecture," *Acta Polytechnica Hungarica*, vol. 20, no. 4, 2023.

[6] Care Quality Commission, *Regulation 17: Good governance - health and social care act 2008 (regulated activities) regulations 2014*, `https://www.cqc.org.uk/guidance-regulation/providers/regulations-service-providers-and-managers/health-social-care-act/regulation-17`, Accessed: 23 April 2026, 2014.

[7] L. Z. Rubenstein, "Falls in older people: Epidemiology, risk factors and strategies for prevention," *Age and ageing*, vol. 35, no. suppl_2, pp. ii37–ii41, 2006.

[8] M. A. Aud, "Dangerous wandering: Elopements of older adults with dementia from long-term care facilities," *American Journal of Alzheimer's Disease & Other Dementias®*, vol. 19, no. 6, pp. 361–368, 2004.

Appendix

A UI

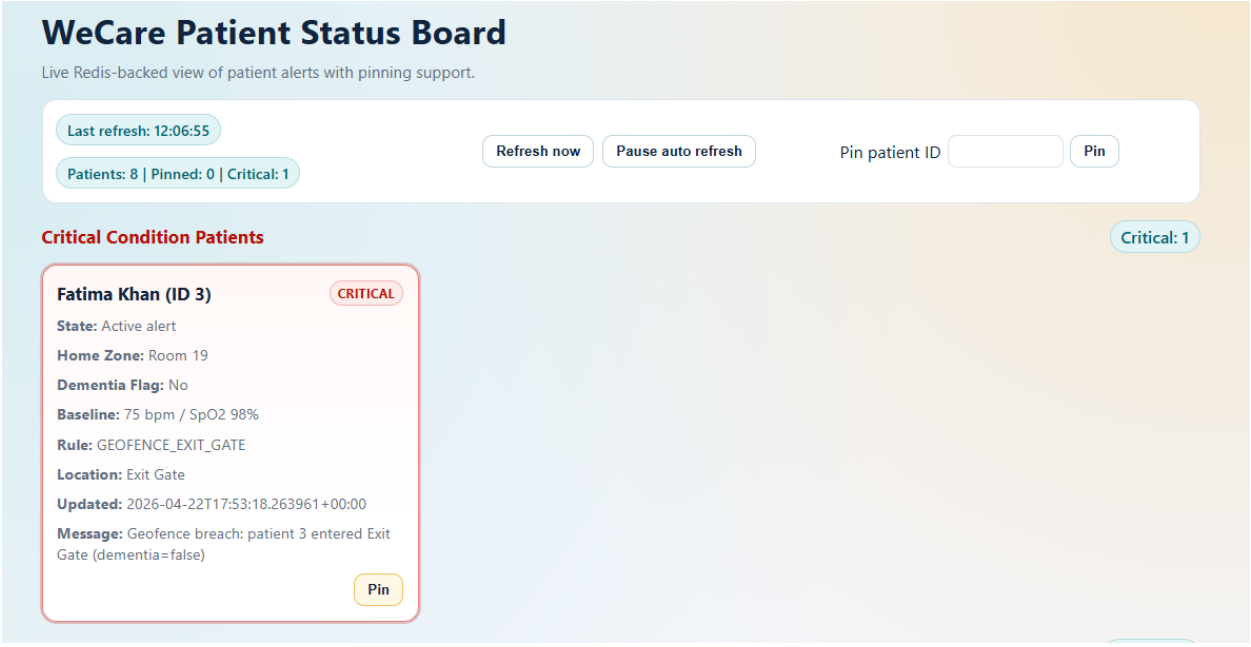

Figure 2: Critical Condition Patients

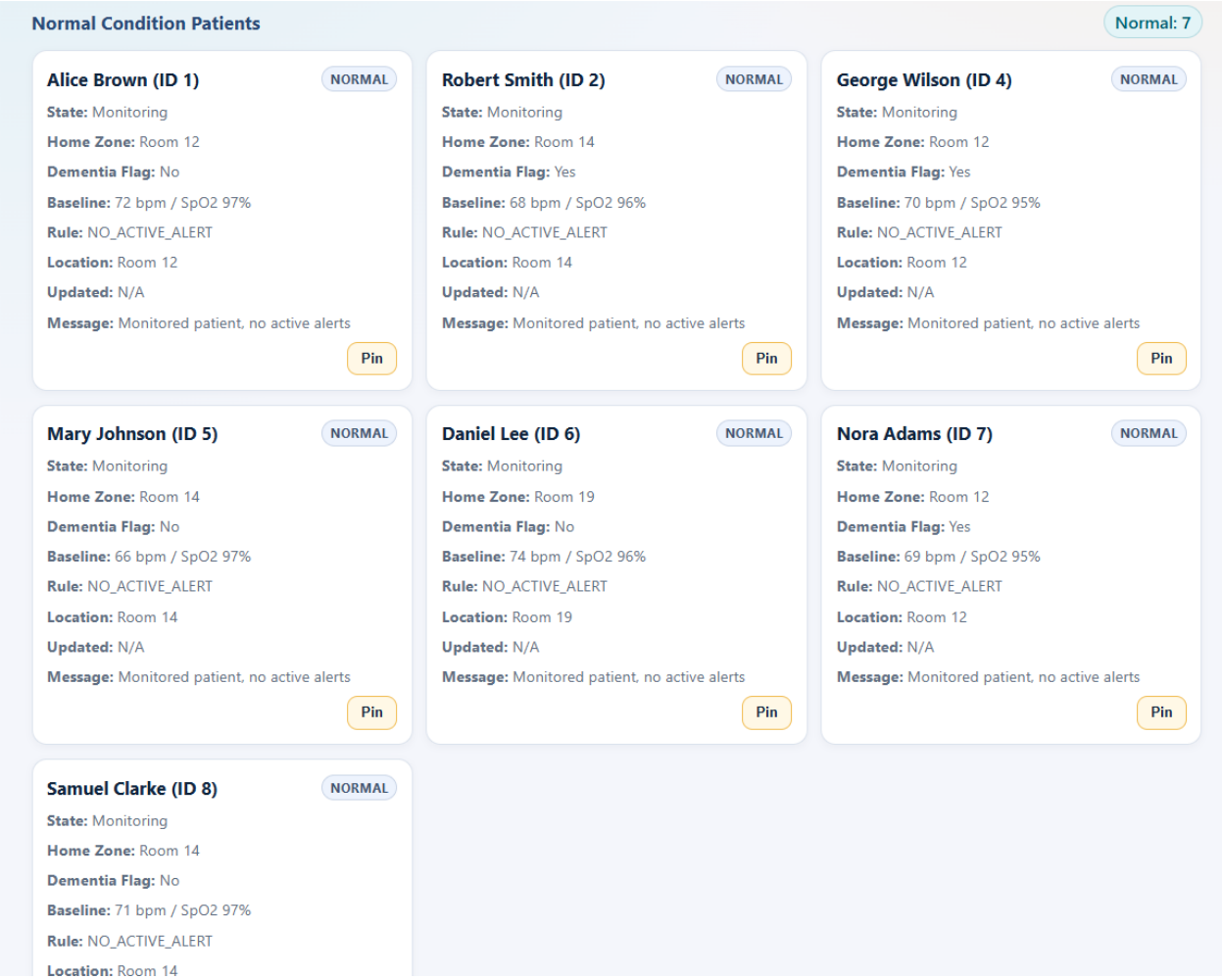

Figure 3: Normal Condition Patients

B RabbitMQ

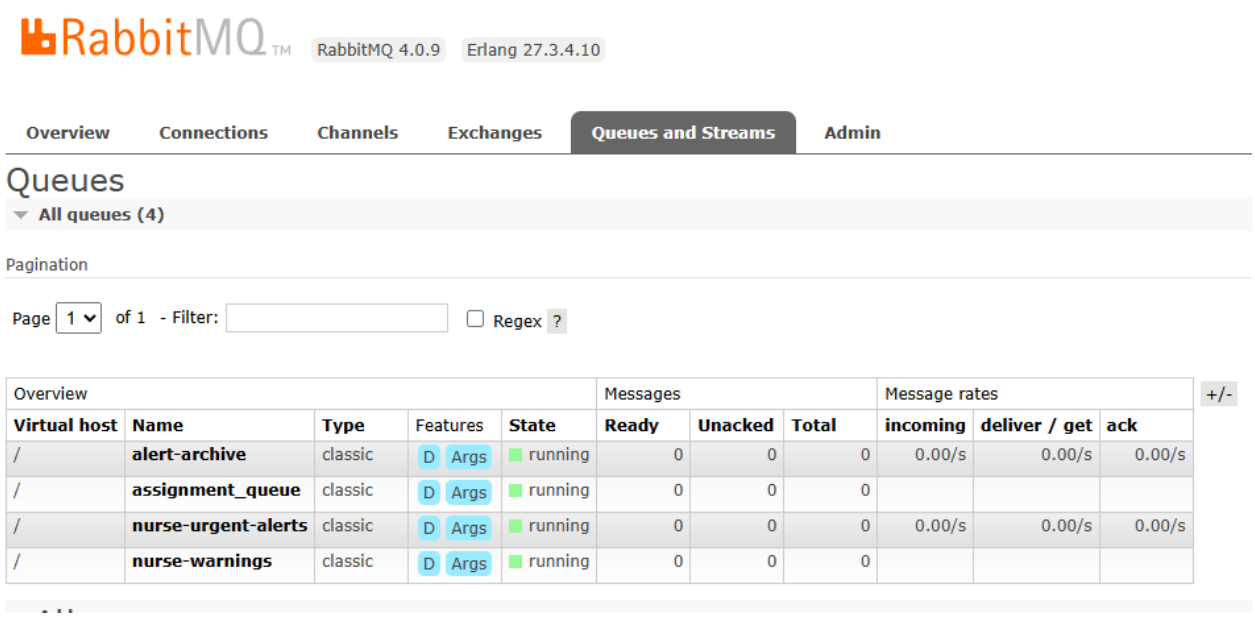

Figure 4: Rabbit Consumer Queues

C Redis

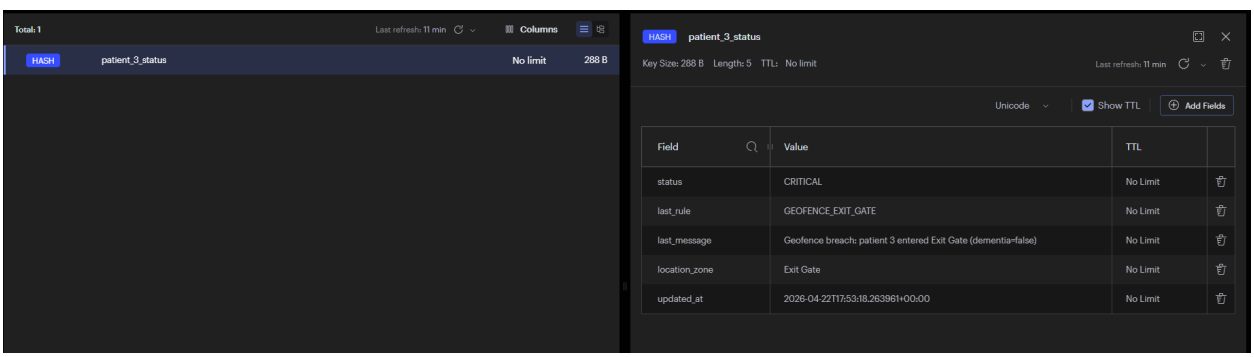

Figure 5: Redis Alarm Data

D DynamoDB Content

The following JSON payload demonstrates the data structure of an archived alert as it is stored in DynamoDB, ensuring a reliable audit trail.

```
{
    "Items": [
        {
            "severity": {
```

```
            "S": "CRITICAL"
        },
        "archived_at": {
            "S": "2026-04-23T11:47:42.034747100Z"
        },
        "parse_error": {
            "S": "false"
        },
        "rule": {
            "S": "FALL_DETECTED"
        },
        "message": {
            "S": "Fall detected for patient 3 in Room 19"
        },
        "detected_at": {
            "S": "2026-04-23T11:47:41.841645200Z"
        },
        "raw_payload": {
            "S": "{\"severity\":\"CRITICAL\",\"incident_started_at
                ↪ \":\"2026-04-23T11:47:41.836Z\",\"dedup_key\":\"3|
                ↪ FALL_DETECTED|ROOM_19|291355ab-705f-4db6-932d-
                ↪ c7c1c98b289a\",\"rule\":\"FALL_DETECTED\",\"
                ↪ heart_rate\":69,\"message\":\"Fall detected for
                ↪ patient 3 in Room 19\",\"incident_event_type\":\"
                ↪ INCIDENT_OPEN\",\"detected_at\":\"2026-04-23T11
                ↪ :47:41.841645200Z\",\"telemetry_timestamp
                ↪ \":\"2026-04-23T11:47:41.670449+00:00\",\"
                ↪ incident_id\":\"291355ab-705f-4db6-932d-c7c1c98b289a
                ↪ \",\"patient_id\":3,\"alert_id\":\"22fdb12d-768c-43
                ↪ f9-ad0e-e6972e09016d\",\"dementia_flag\":false,\"
                ↪ location_zone\":\"Room 19\",\"movement_status\":\"
                ↪ Fall Detected\",\"incident_key\":\"3|FALL_DETECTED|
                ↪ ROOM_19\",\"active_incident_count\":1,\"oxygen_level
                ↪ \":98}"
        },
        "telemetry_timestamp": {
            "S": "2026-04-23T11:47:41.670449+00:00"
        },
        "patient_id": {
            "N": "3"
        },
        "alert_id": {
            "S": "22fdb12d-768c-43f9-ad0e-e6972e09016d"
        },
        "location_zone": {
            "S": "Room 19"
        },
        "movement_status": {
            "S": "Fall Detected"
        },
        "source_queue": {
            "S": "alert-archive"
        }
    }
```

```
    ],
    "Count": 1,
    "ScannedCount": 1,
    "ConsumedCapacity": null
}
```


To demonstrate the full lifecycle of an alert, the following payload shows how the system records an incident's resolution. Notice the shift in `severity` to `NORMAL` and the updated `movement_status`.


```
{
    "Items": [
        {
            "severity": {
                "S": "NORMAL"
            },
            "archived_at": {
                "S": "2026-04-23T11:49:48.319991200Z"
            },
            "parse_error": {
                "S": "false"
            },
            "rule": {
                "S": "FALL_DETECTED"
            },
            "message": {
                "S": "Patient recovered. Fall incident resolved."
            },
            "detected_at": {
                "S": "2026-04-23T11:49:48.282655500Z"
            },
            "raw_payload": {
                "S": "{\"severity\":\"NORMAL\",\"incident_started_at
                    ↪ \":\"2026-04-23T11:47:41.836Z\",\"dedup_key\":\"3|
                    ↪ FALL_DETECTED|ROOM_19|291355ab-705f-4db6-932d-
                    ↪ c7c1c98b289a\",\"rule\":\"FALL_DETECTED\",\"
                    ↪ heart_rate\":81,\"message\":\"Patient recovered.
                    ↪ Fall incident resolved.\",\"incident_event_type\":\"
                    ↪ INCIDENT_RESOLVED\",\"detected_at\":\"2026-04-23T11
                    ↪ :49:48.282655500Z\",\"telemetry_timestamp
                    ↪ \":\"2026-04-23T11:49:48.177965+00:00\",\"
                    ↪ incident_id\":\"291355ab-705f-4db6-932d-c7c1c98b289a
                    ↪ \",\"patient_id\":3,\"alert_id\":\"b6e19bed-add4-40
                    ↪ d7-8ba0-d72066f8ae39\",\"dementia_flag\":false,\"
                    ↪ location_zone\":\"Room 19\",\"movement_status\":\"
                    ↪ Recovered\",\"incident_key\":\"3|FALL_DETECTED|
                    ↪ ROOM_19\",\"active_incident_count\":0,\"oxygen_level
                    ↪ \":97}"
            },
            "telemetry_timestamp": {
                "S": "2026-04-23T11:49:48.177965+00:00"
            },
            "patient_id": {
                "N": "3"
```

```
            },
            "alert_id": {
                "S": "b6e19bed-add4-40d7-8ba0-d72066f8ae39"
            },
            "location_zone": {
                "S": "Room 19"
            },
            "movement_status": {
                "S": "Recovered"
            },
            "source_queue": {
                "S": "alert-archive"
            }
        }
    ]
}
```